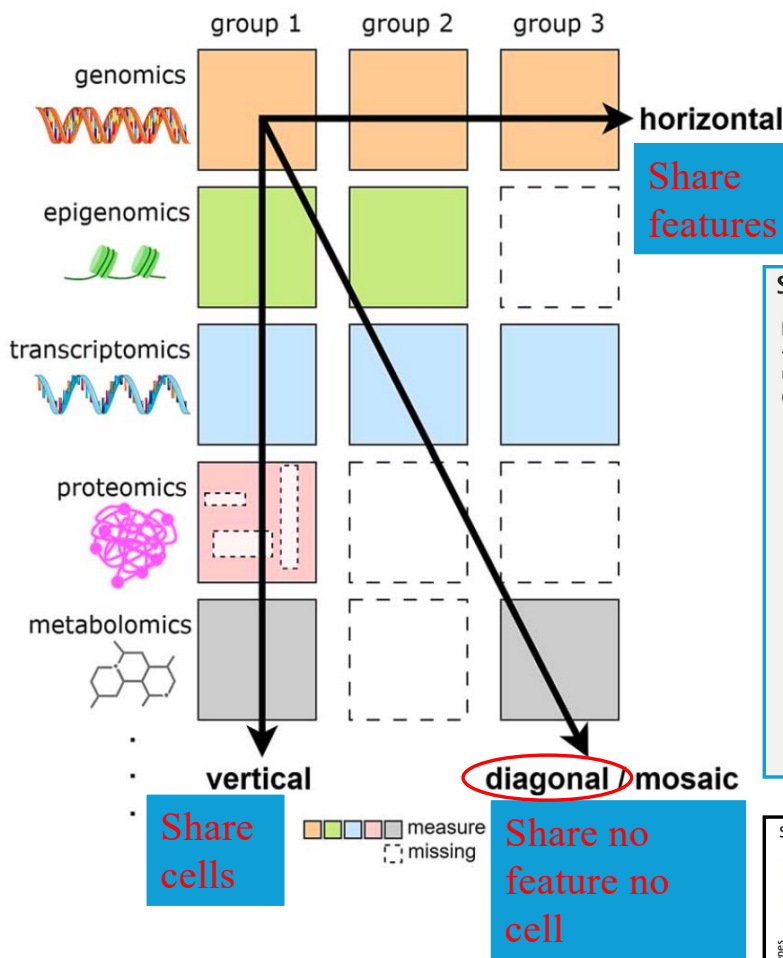




Pure diagonal integration:

- * Share no feature no cell
- * **Topological alignment**
- * Assume similar/shared celltype structures
- * May perform poorly
- * Make more sense for spatial data (PASTE2, STalign)
- * e.g. MMD-MA, PAMONA, SCOT, UniCom.

SCOT works in three steps:

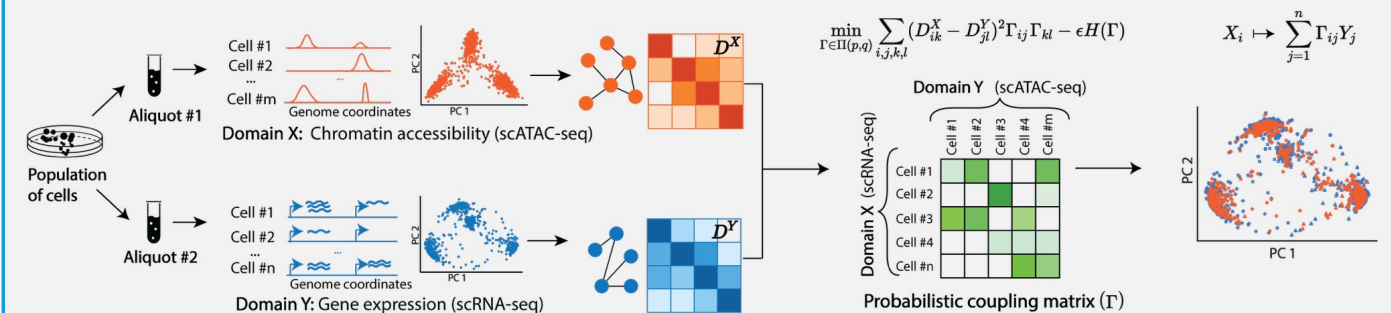
INPUT: Two different single-cell -omics datasets that share some underlying biological manifold (e.g. from the same population)

STEP 1. Computing intra-domain distance matrices by constructing a kNN graph per domain based on correlations and computing shortest distances

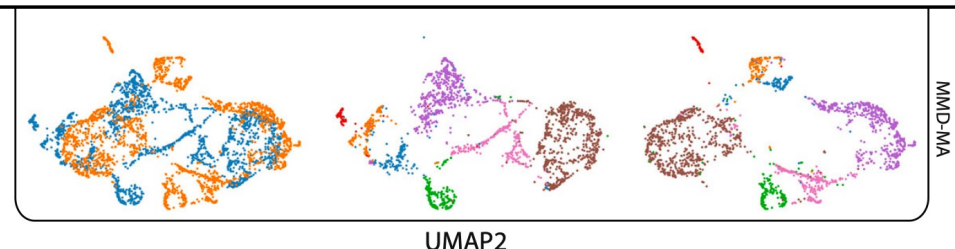
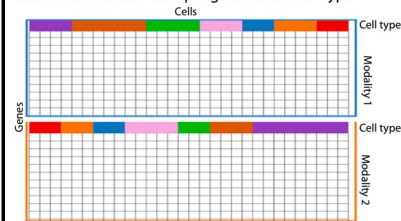
STEP 2. Finding cell-to-cell correspondence probabilities via Gromov-Wasserstein optimal transport with entropic regularization, given D^X , D^Y , and marginal probabilities p , q (uniform distributions defined over the data points in each domain)

STEP 3: Alignment via barycentric projection

OUTPUT: Aligned datasets



Scenario 5: Random sampling from each cell-type



Prior knowledge guidance: Mosaic integration

- **Linked features** (partially share features – horizontal integration)
 - ❖ RNA+ATAC (strong)
 - ❖ RNA+surfaceProtein (weak)
 - **Use only linked features:**
 - E.g. **Seurat_CCA/rPCA**, **LIGER-iNMF**
 - **Linked+unlinked features:**
 - E.g. **scMODEL**, **GLUE** ,
 - **scVI**, **scANVI**, **MultiVI**, **LIGER-UINMF**
- **Leverage joint-profiling** (partially share cells – vertical integration)
 - E.g. **MIDAS**, **Seurat-bridge** , **Stabmap**,
 - **MultiVI**, **Cobolt**, **Multigrade**, **scMoMAT**, **scVAEit**
- **Leverage similar histology in spatial multi-omics**
 - **spatialEx+**

Linear
NMF
Autoencoder

Linked features

scRNA+scATAC

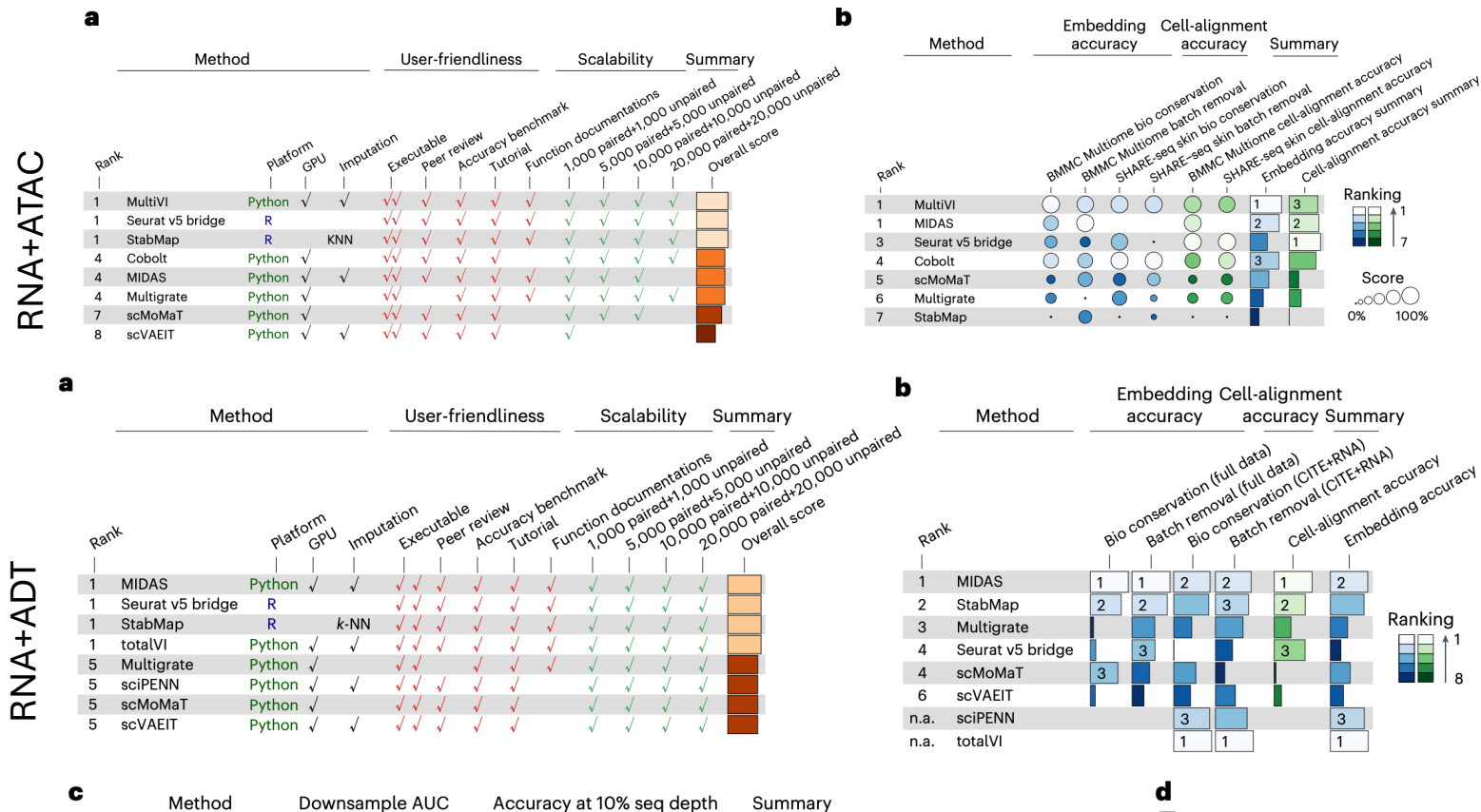
- scATAC summarized to gene activity
- Regulatory interactions: feature correspondence

a

Method			User-friendliness					Scalability						Summary		
Rank		Platform	GPU	Executable	Peer review	Accuracy benchmark	Tutorial	Function documentations	500	1,000	5,000	10,000	50,000	100,000	500,000	Overall score
1	GLUE	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	LIGER iNMF	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	LIGER online iNMF	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	LIGER UINMF	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	MaxFuse	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	SCALEX	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	Seurat v3 CCA	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	SIMBA	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
2	uniPort	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
10	bindSC	R		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
10	MultiMAP	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
12	Pamona	Python		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
13	unionCom	Python	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	
14	GCN-SC	Python	✓	✓												

Fu et al. 2025 Nature methods. <https://doi.org/10.1038/s41592-025-02737-9>

Leverage joint-profiling



Fu et al. 2025 Nature methods. <https://doi.org/10.1038/s41592-025-02737-9>